

Research Notes on Explainable artificial intelligence

Explainable artificial intelligence

>> Section 1

Understanding versus trust The goal of explainability to end users of AI systems is to increase trust in the systems, even "address concerns about lack of 'fairness' and discriminatory effects". However, even with a good understanding of an AI system, end users may not necessarily trust the system. In one study, participants were presented with combinations of white-box and black-box explanations, and static and interactive explanations of AI systems. While these explanations served to increase both their self-reported and objective understanding, it had no impact on their level of trust, which remained skeptical. This outcome was especially true for decisions that impacted the end user in a significant way, such as graduate school admissions. Participants judged algorithms to be too inflexible and unforgiving in comparison to human decision-makers; instead of rigidly adhering to a set of rules, humans are able to consider exceptional cases as well as appeals to their initial decision. For such decisions, explainability will not necessarily cause end users to accept the use of decision-making algorithms. We will need to either turn to another method to increase trust and acceptance of decision-making algorithms, or question the need to rely solely on AI for such impactful decisions in the first place. However, some emphasize that the purpose of explainability of artificial intelligence is not to merely increase users' trust in the system's decisions, but to calibrate the users' level of trust to the correct level. According to this principle, too much or too little user trust in the AI system will harm the overall performance of the human-system unit. When the trust is excessive, the users are not critical of possible mistakes of the system and when the users do not have enough trust in the system, they will not exhaust the benefits inherent in it.

>> Section 2

Voting Cailloux and Endriss present a method for explaining voting rules using the axioms that characterize them. They exemplify their method on the Borda voting rule. Peters, Procaccia, Psomas and Zhou present an algorithm for explaining the outcomes of the Borda rule using $O(m^2)$ explanations, and prove that this is tight in the worst case.

>> Section 3

A model is transparent "if the processes that extract model parameters from training data and generate labels from testing data can be described and motivated by the approach designer." Interpretability describes the possibility of comprehending the ML model and presenting the underlying basis for decision-making in a way that is

understandable to humans. Explainability is a concept that is recognized as important, but a consensus definition is not yet available; one possibility is "the collection of features of the interpretable domain that have contributed, for a given example, to producing a decision (e.g., classification or regression)". In summary, Interpretability refers to the user's ability to understand model outputs, while Model Transparency includes Simulatability (reproducibility of predictions), Decomposability (intuitive explanations for parameters), and Algorithmic Transparency (explaining how algorithms work). Model Functionality focuses on textual descriptions, visualization, and local explanations, which clarify specific outputs or instances rather than entire models. All these concepts aim to enhance the comprehensibility and usability of AI systems. If algorithms fulfill these principles, they provide a basis for justifying decisions, tracking them and thereby verifying them, improving the algorithms, and exploring new facts. Sometimes it is also possible to achieve a high-accuracy result with white-box ML algorithms. These algorithms have an interpretable structure that can be used to explain predictions. Concept Bottleneck Models, which use concept-level abstractions to explain model reasoning, are examples of this and can be applied in both image and text prediction tasks. This is especially important in domains like medicine, defense, finance, and law, where it is crucial to understand decisions and build trust in the algorithms. Many researchers argue that, at least for supervised machine learning, the way forward is symbolic regression, where the algorithm searches the space of mathematical expressions to find the model that best fits a given dataset. AI systems optimize behavior to satisfy a mathematically specified goal system chosen by the system designers, such as the command "maximize the accuracy of assessing how positive film reviews are in the test dataset." The AI may learn useful general rules from the test set, such as "reviews containing the word 'horrible' are likely to be negative." However, it may also learn inappropriate rules, such as "reviews containing 'Daniel Day-Lewis' are usually positive"; such rules may be undesirable if they are likely to fail to generalize outside the training set, or if people consider the rule to be "cheating" or "unfair." A human can audit rules in an XAI to get an idea of how likely the system is to generalize to future real-world data outside the test set.

>> Section 4

Adversarial parties By making an AI system more explainable, we also reveal more of its inner workings. For example, the explainability method of feature importance identifies features or variables that are most important in determining the model's output, while the influential samples method identifies the training samples that are

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most influential in determining the output, given a particular input. Adversarial parties could take advantage of this knowledge. For example, competitor firms could replicate aspects of the original AI system in their own product, thus reducing competitive advantage. An explainable AI system is also susceptible to being "gamed"—influenced in a way that undermines its intended purpose. One study gives the example of a predictive policing system; in this case, those who could potentially "game" the system are the criminals subject to the system's decisions. In this study, developers of the system discussed the issue of criminal gangs looking to illegally obtain passports, and they expressed concerns that, if given an idea of what factors might trigger an alert in the passport application process, those gangs would be able to "send guinea pigs" to test those triggers, eventually finding a loophole that would allow them to "reliably get passports from under the noses of the authorities".

>> Section 5

Goals Cooperation between agents – in this case, algorithms and humans – depends on trust. If humans are to accept algorithmic prescriptions, they need to trust them. Incompleteness in formal trust criteria is a barrier to optimization. Transparency, interpretability, and explainability are intermediate goals on the road to these more comprehensive trust criteria. This is particularly relevant in medicine, especially with clinical decision support systems (CDSS), in which medical professionals should be able to understand how and why a machine-based decision was made in order to trust the decision and augment their decision-making process. AI systems sometimes learn undesirable tricks that do an optimal job of satisfying explicit pre-programmed goals on the training data but do not reflect the more nuanced implicit desires of the human system designers or the full complexity of the domain data. For example, a 2017 system tasked with image recognition learned to "cheat" by looking for a copyright tag that happened to be associated with horse pictures rather than learning how to tell if a horse was actually pictured. In another 2017 system, a supervised learning AI tasked with grasping items in a virtual world learned to cheat by placing its manipulator between the object and the viewer in a way such that it falsely appeared to be grasping the object. One transparency project, the DARPA XAI program, aims to produce "glass box" models that are explainable to a "human-in-the-loop" without greatly sacrificing AI performance. Human users of such a system can understand the AI's cognition (both in real-time and after the fact) and can determine whether to trust the AI. Other applications of XAI are knowledge extraction from black-box models and model comparisons. In the context

of monitoring systems for ethical and socio-legal compliance, the term "glass box" is commonly used to refer to tools that track the inputs and outputs of the system in question, and provide value-based explanations for their behavior. These tools aim to ensure that the system operates in accordance with ethical and legal standards, and that its decision-making processes are transparent and accountable. The term "glass box" is often used in contrast to "black box" systems, which lack transparency and can be more difficult to monitor and regulate. The term is also used to name a voice assistant that produces counterfactual statements as explanations.